AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2014

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

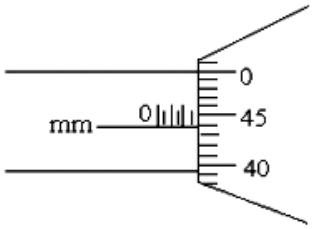
4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1.

(Total 3 Marks)

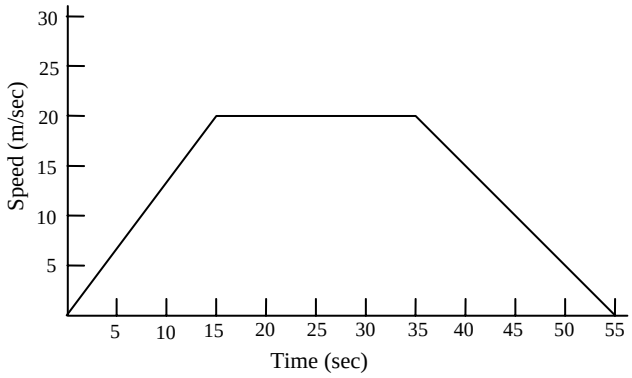
The given diagram shows the diameter of a copper wire measured by a screw gauge.



With the help of the reading shown in the diagram, calculate the diameter of the copper wire in scientific notation.

Q.2. (Total 6 Marks)

a. A speed-time graph for a car travelling on a straight road is shown below. (3 Marks)



i. What is the distance travelled during the first 15 sec?
ii. What is the deceleration during the last 10 sec?
iii. What is the average speed of the whole journey?

b. A paratrooper jumps from a helicopter. He does not open his parachute straight away.

The table shows his speed during the 10 seconds after his jump. (3 Marks)

Time in sec	0	1	2	3	4	5	6	7	8	9	10
Speed in m/s	0	10	20	30	40	50	35	27	22	20	20

i. After how many seconds does he open his parachute?

_____

ii. Name the force that pulls the paratrooper down.

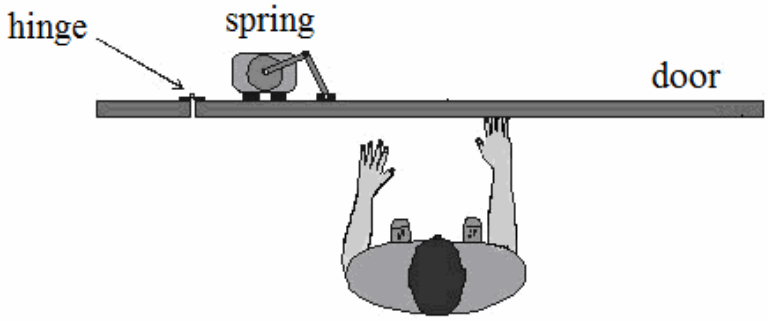
_____

iii. Name the force that acts in an upward direction.

_____

Q.3. (Total 3 Marks)

The diagram below shows a person who is about to open the door.



a. Suggest any TWO changes that the person can make in order to open the door more easily.

(2 Marks)

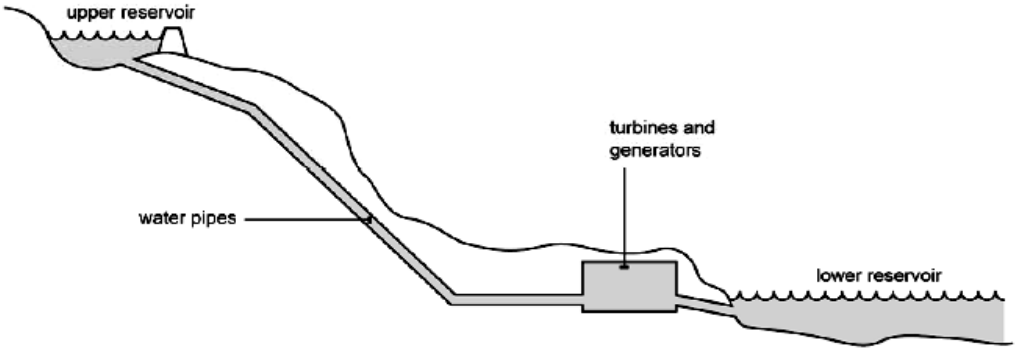
b. At what angle does the turning effect have maximum value? (1 Mark)

Q.4. (Total 3 Marks)

Is Newton’s law of gravitation consistent with Newton's third law of motion? Give any TWO conditions of the third law of motion.

Q.5. (Total 6 Marks)

a. In a hydroelectric power generation station, energy transformations take place. Describe the process of power generation by hydroelectric dam and conversion of different forms of energy into electrical energy. (3 Marks)



b. State any THREE disadvantages of hydroelectric power generation. (3 Marks)

Q.6. (Total 4 Marks)

a. Why does evaporation of water cause cooling? (2 Marks)

_____

_____

_____

_____

b. Why are conduction and convection not involved in the transfer of heat from the sun to the earth? (2 Marks)

_____

_____

_____

_____

Q.7. (Total 5 Marks)

EITHER

a. Derive Newton's second equation of motion for a body moving with initial velocity V_i with constant acceleration a in time t.

OR

b. The following diagram represents balloons suspended in the air.



Describe why balloons remain suspended in the air. Name TWO forces which are illustrated in the above diagram.

Q.8. (Total 5 Marks)

EITHER

a. Describe the process of power generation from fossil fuel input by drawing a block diagram of the process.

OR

b. The pressure in a pressure cooker is raised by squeezing more gas into a limited space. Explain why gases exert greater pressure when they are squeezed into smaller space.

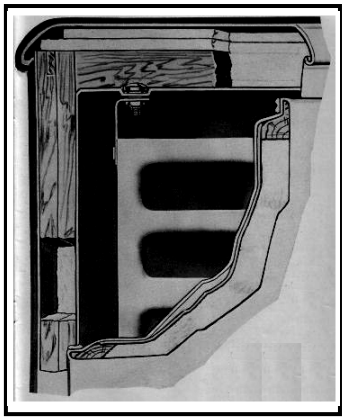
Q.9. (Total 5 Marks)

EITHER

a. Rise in temperature of a body means an increase in its internal energy. Explain the given statement.

OR

b. The given diagram shows a cross-section of the walls of a refrigerator. Explain why the walls are filled with fibre glass and lined with crinkled aluminium.



END OF PAPER

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